5. Missionaries and Cannibals Problem

Aim:

To write a Python program to solve the Missionaries and Cannibals problem using a search algorithm.

Algorithm:

1. Start the program.
2. Define the initial state with 3 missionaries and 3 cannibals.
3. Define the boat capacity as 2.
4. Generate possible boat movements.
5. Check that missionaries are not outnumbered by cannibals on either bank.
6. Avoid repeated states.
7. Continue until everyone reaches the opposite bank.
8. Display the solution path.
9. Stop.

Code:

```
def safe(m, c):
    return m == 0 or m >= c

def solve():
    q = deque([(start, [])])
    visited = {start}

    moves = [(1,0), (2,0), (0,1),
             (0,2), (1,1)]

    while q:
        state, path = q.popleft()
        m, c, boat = state

        if state == goal:
            return path + [state]

        for dm, dc in moves:
            if boat == 1:
                nm, nc = m-dm, c-dc
                nb = 0
            else:
                nm, nc = m+dm, c+dc
                nb = 1

            if 0 <= nm <= 3 and 0 <= nc <= 3:
                if safe(nm, nc) and safe(3-nm, 3-nc):
                    new = (nm, nc, nb)

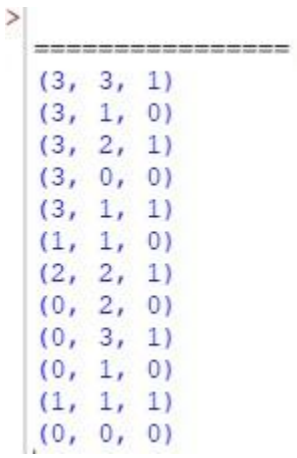
                    if new not in visited:
                        visited.add(new)
                        q.append((new, path+[state]))

    return None

result = solve()

for state in result:
    print(state)
```

Output:



Result:

Thus, the Python program to solve the Missionaries and Cannibals problem was executed successfully.